FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Main)-2020

TEST DATE: 27-08-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

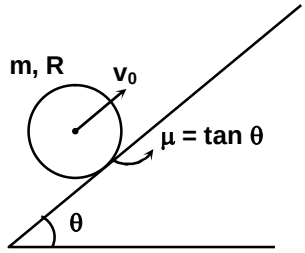
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A solid sphere of mass 'm' and radius 'R' is placed on a rough inclined plane of inclination 'θ' and then imparted an initial velocity 'v_0' up the plane as shown in the figure. The coefficient of friction between the sphere and the inclined plane is $\mu = \tan\theta$. Then the time after which pure rolling motion of the sphere starts will be



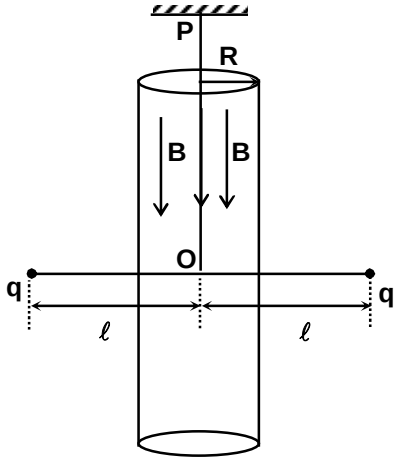
(A) $\frac{v_0}{9g\sin\theta}$

(B) $\frac{2v_0}{9g\sin\theta}$

(C) $\frac{v_0}{3g\sin\theta}$

(D) $\frac{2v_0}{3g\sin\theta}$

2. A uniform magnetic field 'B' is present in a cylindrical region of radius 'R' and parallel to the axis of the cylindrical region as shown in the figure. Two particles each having charge 'q' and mass 'm' are attached to the ends of the rod of length 2ℓ and mass m and this arrangement hangs with the help of a vertical rope PO as shown in the figure. At t = 0, the magnetic field is switched off, then the angular velocity acquired by the rod is (neglect twisting constant of rope)



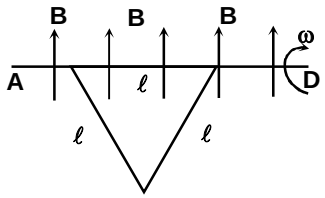
(A) $\frac{3qBR^2}{7m\ell^2}$

(B) $\frac{3qBR^2}{2m\ell^2}$

(C) $\frac{qBR^2}{5m\ell^2}$

(D) $\frac{qBR^2}{4m\ell^2}$

3. A conducting equilateral triangular loop having side length 'ℓ' and resistance 'R' is suspended in a vertical plane from a horizontal axis 'AD' coinciding with one side of the loop as shown in the figure. A uniform vertical upward magnetic field 'B' is existing in the region. If the triangular loop is rotated with a constant angular velocity 'ω' about the horizontal axis 'AD'. Then the average power developed in the loop is



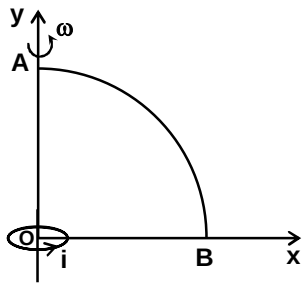
(A) $\frac{3B^2\omega^2\ell^4}{32R}$

(B) $\frac{B^2\omega^2\ell^4}{16R}$

(C) $\frac{B^2\omega^2\ell^4}{32R}$

(D) $\frac{3B^2\omega^2\ell^4}{16R}$

4. Consider a quarter circular conducting ring of large radius 'R' located in xy-plane with its centre at origin and a small circular loop of radius 'a' carrying current 'i' is placed in xz-plane with centre at origin as shown in the figure. If the quarter circular ring is rotated with a constant angular velocity 'ω' about the y-axis, then the emf induced between its ends A and B is



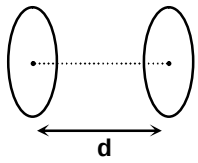
(A) $\frac{3\mu_0a^2\omega i}{4R}$

(B) $\frac{\mu_0a^2\omega i}{4R}$

(C) $\frac{\mu_0a^2\omega i}{8R}$

(D) $\frac{3\mu_0a^2\omega i}{8R}$

5. Two small circular current carrying loops each having radius 'R' and current 'I' in the same sense are placed coaxially at a distance 'd' apart. Then the force of attraction between the two current loops is
(assume d >>R)



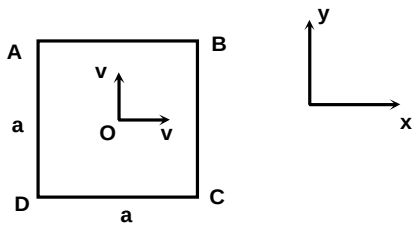
(A) $I\pi R^2\left(\frac{\mu_0 IR^2}{2d^4}\right)$

(B) $I\pi R^2\left(\frac{5\mu_0 IR^2}{2d^4}\right)$

(C) $I\pi R^2\left(\frac{3\mu_0 IR^2}{2d^4}\right)$

(D) $I\pi R^2\left(\frac{7\mu_0 IR^2}{2d^4}\right)$

6. A uniform square lamina with side 'a' is moving with a constant velocity $\left(v\hat{i}+v\hat{j}\right)$ with its plane lying on a smooth horizontal xy plane surface as shown in the figure. The angular velocity of the square lamina just after getting hinged at the vertex 'A' is



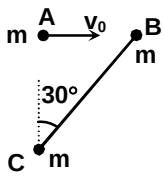
(A) $\frac{v}{2a}$

(B) $\frac{3v}{2a}$

(C) $\frac{5v}{2a}$

(D) $\frac{3v}{4a}$

7. Two identical balls B and C of same mass 'm' are connected by a massless inextensible string and placed on a smooth horizontal surface as shown in the figure. A third ball A of same mass 'm' moving with a velocity v_0 collides with the ball 'B' and after collision the ball A keeps moving in the same direction with a velocity $\frac{v_0}{5}$, then the impulse applied by the string on the ball C during collision is



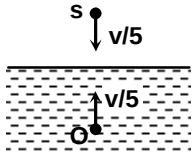
(A) mv_0

(B) $\frac{mv_0}{2}$

(C) $\frac{mv_0}{3}$

(D) $\frac{mv_0}{5}$

8. In the figure shown, a source 'S' is moving in air with a constant velocity $\frac{v}{5}$ and an observer 'O' is moving with a constant velocity $\frac{v}{5}$ in water. The source emits sound of frequency f. The velocity of sound in air is v and that in water 4v. Then the wavelength of sound received by the observer is



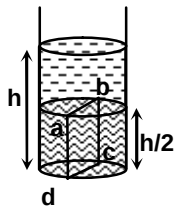
(A) $\frac{8v}{5f}$

(B) $\frac{16v}{5f}$

(C) $\frac{24v}{5f}$

(D) $\frac{32v}{5f}$

9. Liquid having density 'ρ' is filled in a beaker of radius R upto height h as shown in the figure. Then the force on the section abcd due to the liquid in the right side is



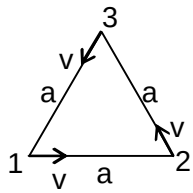
(A) $\left(P_0+\rho g\frac{3h}{4}\right)2R\times\frac{h}{3}$

(B) $\left(P_0+\rho g\frac{3h}{4}\right)Rh$

(C) $\left(P_0+\rho g\frac{3h}{4}\right)2Rh$

(D) $\left(P_0+\rho g\frac{3h}{4}\right)R\times\frac{h}{2}$

10. Three particles are located at the vertices of an equilateral triangle of side 'a'. They all start moving simultaneously with the same constant speed 'v' with the 1st particle always heading for the 2nd, the 2nd particle always heading for the 3rd and 3rd is always heading for the 1st. Then the acceleration of each particle when the separation between the particles becomes 'a/2' will be



(A) $\frac{v^2}{2a}$

(B) $\frac{v^2}{a}$

(C) $\frac{\sqrt{3}v^2}{2a}$

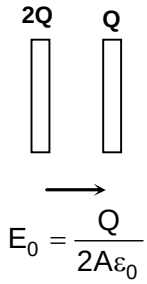
(D) $\frac{\sqrt{3}v^2}{a}$

11. A fire cracker when bursted produces 80 dB sound at a distance of 10 cm. At what distance from the fire cracker should a person stand so that sound level heard by him is 40 dB? (assume sound energy is spreaded equally in all directions)
(A) 5 m
(B) 10 m
(C) 15 m
(D) 20 m

12. A cylindrical wire of copper having area of cross section 5 mm^2 is carrying a current of 1A. The wire is stretched so that its area of cross section becomes 2.5 mm^2. How does the mobility of charge carrier will change, if a current of 4A is passed through the wire.
(A) remain same
(B) increase
(C) decrease
(D) can't say

13. A sonometer wire vibrating in first overtone has length of 17 cm. Tension in the wire is 289 N and mass per unit length is 10 gm/cm. When it is sounded together with a tuning fork, a beats frequency of 5 Hz is observed. If tension in the wire is slightly increased, no beats are heard. What is the possible frequency of the tuning fork?
(A) 50 Hz
(B) 75 Hz
(C) 90 Hz
(D) 105 HZ

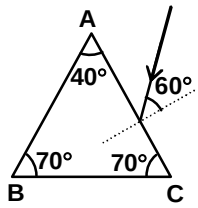
14. Two large parallel conducting plates of charges 2Q and Q are placed parallel to each other at a small separation 'd' in a uniform external electric field $E_0 = \frac{Q}{2A\varepsilon_0}$ perpendicular to the plates as shown. The surface area of each plate is A. The charge on the left surface of the left plate and right surface of the right plate are respectively.



(A) 2Q and Q
(B) 4Q and Q
(C) Q and 2Q
(D) 5Q and 2Q

15. An unpolarised light is passed through three polaroids 1, 2 and 3 respectively. Angle between the transmission axes of the polaroids 1 and 2 is 37°. Find the angle between the transmission axes of the polaroids 2 and 3, if the intensity of the emerging light is 8% of the original intensity of light.
(A) 30°
(B) 60°
(C) 75°
(D) 90°

16. A ray of light is incident on one face of a prism as shown. After refraction through this surface the ray is incident on the base of the prism. The refractive index of the prism is $\sqrt{3}$. The net deviation produced in the ray due to prism is $\left[\text{Take } \sin^{-1}\left(\frac{1}{\sqrt{3}}\right) = 35°\right]$

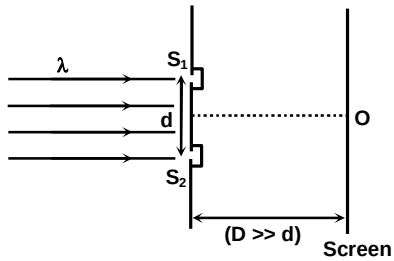


(A) 80° clockwise
(B) 160° clockwise
(C) 180° clockwise
(D) 240° clockwise

17. A speech signal of 5 kHz is used to modulate a carrier signal of frequency 1 MHz using amplitude modulation. The frequency of side bands will be
(A) 995 kHz, 1005 kHz
(B) 895 kHz, 5 kHz
(C) 695 kHz, 5 kHz
(D) 1995 kHz, 1005 kHz

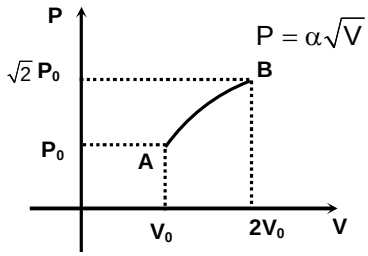
18. A wire of length ℓ = (5 ± 0.05) cm, radius r = (0.6 ± 0.006)cm and density ρ = (0.4 ± 0.004) gm/cm^3 is given. The maximum percentage error in the mass of the wire is
(A) 1
(B) 2
(C) 4
(D) 5

19. In young's double slit experiment, when the upper and lower slits are covered by two thin slabs of refractive indices 1.8 and 1.5 respectively and having same thickness, the central bright fringe is shifted to the third bright fringe of the interference pattern obtained on the screen by using a light of wavelength 5000 Å. Then the thickness of each slab is (D >>d)



(A) 1 μm
(B) 3 μm
(C) 5 μm
(D) 6 μm

20. An ideal diatomic gas is expanded from volume V_0 to $2V_0$ in the process $P = \alpha\sqrt{V}$ (where α is a constant) as shown. Then the molar heat capacity of the gas in the given process is



(A) $\frac{19R}{6}$

(B) $\frac{17R}{6}$

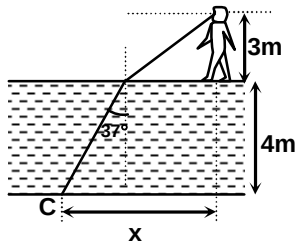
(C) $\frac{21R}{6}$

(D) $\frac{27R}{6}$

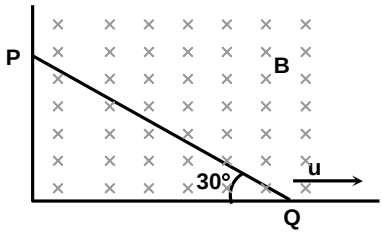
SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A man is standing at the edge of 4 m deep swimming pool, completely filled with water of refractive index μ = (4/3). The eyes of the man are 3m above the edge of the swimming pool. A coin is located at the bottom of the pool. The ray coming from the coin 'C' at an angle 37° from the normal to the water-air interface reaches the eye of the man. Then find the horizontal distance in meter (represented by x in the figure) of the coin from the eye of the man is



22. A conducting rod PQ of length 'ℓ' is resting between a non-conducting wall and the horizontal floor as shown in the figure in the presence of a uniform magnetic field B. The lower end of the rod 'Q' has velocity u at the shown instant. If the potential difference between P and Q at that instant is $\frac{Bu\ell}{n}$. Find the value of n.

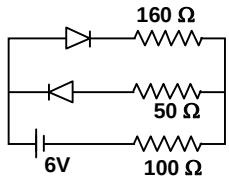


SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. The pitch of a screw gauge is 1 mm and its cap is divided into 100 divisions. There is no zero error in the screw gauge. When a wire is placed between its studs, the main scale reading is 3 mm and 25th division of circular scale coincides with the main scale. Find the diameter (in mm) of the wire.

24. The circuit shown in the figure contains 2 diodes each with a forward resistance of 40 ohms and with infinite backward resistance. If battery voltage is 6V, the current through the 100 ohm resistance (in milliampere) is



25. A uniform circular disc of radius 40 cm is suspended vertically from a point at a distance x from centre of the disc to oscillate under gravity. What should be the value of x (in cm) so that the time period of its small oscillation is minimum? (Take $\sqrt{2} = 1.41$)

Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. Given the following value for the standard molar Gibbs free energy changes. $\Delta G°$, at 25°C:

$Ag^+(aq) + 2NH_3(\ell) \rightarrow Ag(NH_3)_2^+(aq);$

$\Delta G° = -41.0\ kJ\ (mol\ Ag^+)^{-1}$

$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s);$

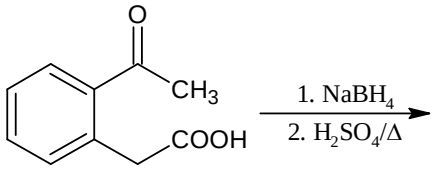
$\Delta G° = -55.6\ kJ\ (mol\ Ag^+)^{-1}$

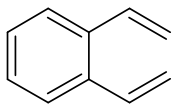
Calculate the value for the thermodynamic equilibrium constant, K, at 25°C for the reaction

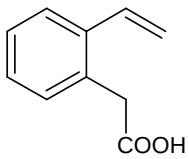
$AgCl(s) + 2NH_3(l) \rightarrow Ag(NH_3)_2^+(aq) + Cl^-(aq)$ (R = 8.3 J K^{-1} mol^{-1})

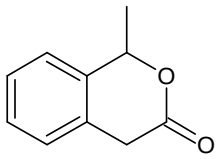
(A) 3.62×10^2
(B) 1.00
(C) 9.94×10^{-1}
(D) 2.76×10^{-3}

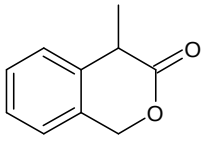
27. Predict the major product of the following reaction.



(A) 

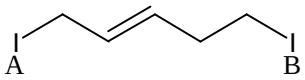
(B) 

(C) 

(D) 

28. Calculate the density of NaCl type ionic solid AB (M_W = 60) in gm/cm^3 if shortest distance between two nearest neighbours is 100 pm & solid having 20% schottky defect ($N_A = 6 \times 10^{23}$)
(A) 10
(B) 20
(C) 30
(D) 40

29. Which one of the two iodine will be more reactive in the S_N1 and S_N2 reaction?



(A) A will be faster in S_N1 reaction but slower in S_N2.
(B) A will be faster both in S_N1 and S_N2 reaction
(C) A and B will be equally reactive
(D) B will be faster in S_N1 and S_N2 reactions

30. Bessemerization is carried out for
I: Fe II: Cu III: Al IV: silver
(A) I, II
(B) II, III
(C) III, IV
(D) I, III

31. The emf of a cell corresponding to the following reaction is 0.199 V at 298 K.

$$Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(0.1\,M) + H_2(g) \quad \left(E^0_{Zn/zn^{+2}} = 0.76\,V\right)$$

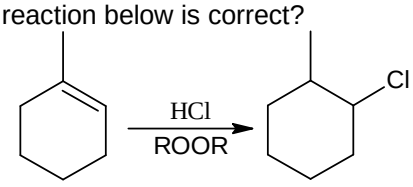
The approximate pH of the solution at the electrode where hydrogen is being produced is $\left(p_{H_2} = 1\,atm\right)$
(A) 8
(B) 9
(C) 10
(D) 11

32. Consider the following reversible reaction:

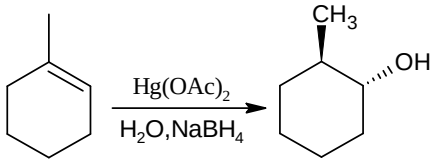
$$NO + NO_3 \rightleftharpoons 2NO_2$$

If 1.0 mol of NO is mixed with 3.0 mol of NO_3, 'x' mol of NO_2 is produced at equilibrium. If 2.0 mol of NO is added further. 'x' mole of NO_2 is further produced. What is the value of equilibrium constant?
(A) 1
(B) 2
(C) 3
(D) 4

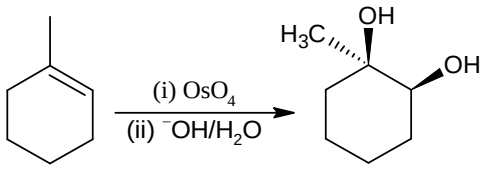
33. Which reaction below is correct?
(A)



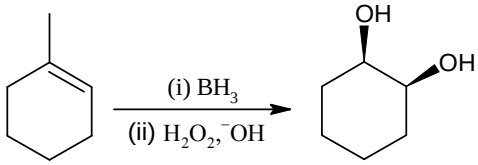
(B)



(C)



(D)



34. Correct order of C–O bond length is:

(A) $CO^{+} > C_2O_4^{2-} > CO$

(B) $CO^{+} > CO > C_2O_4^{2-}$

(C) $C_2O_4^{2-} > CO^{+} > CO$

(D) $C_2O_4^{2-} > CO > CO^{+}$

35. The mass of gas molecule Y is twice the mass of gas molecule 'Z'. The rms speed of 'Y' is twice the rms speed of 'Z' and their respective pressure is 'P_Y' and 'P_Z'. If two samples of 'Y' and "Z' contains same no. of molecules, what will be the ratio of respective pressure of two sample in separate container of equal volume ?

(A) 2
(B) 4
(C) 6
(D) 8

36. Which of the following will not give positive chromyl chloride test?

(A) $CaCl_2$
(B) $HgCl_2$
(C) $ZnCl_2$
(D) $C_6H_5NH_2.HCl$

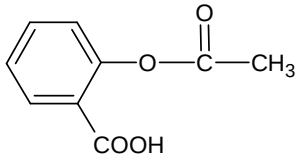
37. Which of the following is not a disaccharide?

(A) Sucrose
(B) Mannose
(C) Lactose
(D) Maltose

38. Graph between $\log\frac{x}{m}$ and $\log P$ is a straight line inclined at an angle θ = 45°. When pressure is 0.5 atm and log *k* = 0.699, the amount of solution adsorbed per g of adsorbent will be:

(A) 2.5 g/g adsorbent
(B) 0.25 g/g adsorbent
(C) 1 g/g adsorbent
(D) 1.5 g/g adsorbent

39. The following compound is used as



(A) An anti-inflammatory compound
(B) Analgesic
(C) Hypnotic
(D) Antiseptic

40. Benzoic acid and carbolic acid can be distinguished by
(A) Aqueous $NaHCO_3$
(B) 2, 4-DNP test
(C) Aqueous NaOH
(D) Aqueous NH_3

41. Hydrolysis of one mole of peroxodi-sulphuric acid produces:
(A) two moles of sulphuric acid
(B) two moles of peroxomono-sulphuric acid
(C) one mole of sulphuric acid, one mole of peroxomono-sulphuric acid
(D) one mole of sulphuric acid, one mole of peroxomono-sulphuric acid and one mole of hydrogen peroxide.

42. Which of the following complex is paramagnetic?
(A) $K_2[Ni(CN)_4]$
(B) $K_4[Fe(CN)_6]$
(C) $[Co(NH_3)_6]Cl_3$
(D) $[Cr(H_2O)_6]SO_4$

43. Industrial preparation of nitric acid by Ostwald's process involves:
(A) Oxidation of NH_3
(B) Reduction of NH_3
(C) Hydrogenation of NH_3
(D) Hydrolysis of NH_3

44. Two samples A and B of an ideal gas, initially at same temperature and pressure, are compressed from volume V to V/2, isothermally for A and adiabatically for B. The final pressure of A will be:
(A) greater than that of B
(B) less than that of B
(C) twice that of B
(D) equal to that of B

45. The dipole moment of H_2O_2 is more than that of H_2O but H_2O_2 is not a good solvent because:
(A) It has a very high dielectric constant so that ionic compounds cannot be dissolved in it
(B) It does not act as an oxidising agent
(C) It acts only as a reducing agent
(D) It dissociates easily and acts as an oxidising agent in chemical reactions.

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. How many geometrical isomers are possible in square planar complex, $[Pt(NH_3)\ (H_2O)\ (Py)\ (NO_2)]^+$

47. In borax molecule $Na_2(B_4O_5(OH)_x]\ .\ YH_2O$ what will be the value of 'x', If x molecules are part of its structural composition.

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. 3129 kJ of heat is liberated when 60 grams of ethane is completely burnt in oxygen. ΔH^0_f for CO_2 and $H_2O(\ell)$ are $-395kJ$ and $-286kJ$ respectively. The heat of formation of ethane is

49. At 18°C the molar conductivities at infinite dilution of NH_4Cl, NaOH& NaCl are 130, 220 & 100 Scm^2mol^{-1} respectively. If the molar conductivity of N/100 solution of NH_4OH is 10 Scm^2mol^{-1}, then find the pH of the NH_4OH solution. (log2 = 0.30)

50. If the osmotic pressure of 0.1M monobasic acid in a solution of pH value '2' at 25^0C is x atm, then what will be x?

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. The tangent at any point P(5, 2) on the parabola $y^2 = 2(x - 3)$ meets the directrix of this parabola at Q then if circumcentre of ΔSPQ where S is focus of parabola is (α, β), then $\alpha + 2\beta$ is equal to
(A) $\frac{13}{2}$
(B) $\frac{11}{2}$
(C) 7
(D) none of these

52. Let $f : R \to R$ be continuous function satisfying $f(x) + f(x + 1) = x^2 + 2x + 3$, then $\int_0^2 f(x)dx$ is equal to
(A) $\frac{38}{3}$
(B) $\frac{32}{3}$
(C) $\frac{13}{3}$
(D) $\frac{16}{3}$

53. From the digits 0, 1, 2, 3, 4, how may 4 digit number can be formed greater than 2100
(A) 250
(B) 249
(C) 350
(D) 349

54. $\lim\limits_{x\to\infty} \frac{30 + 3\sqrt{x} + x^{\frac{1}{3}}}{3 + \sqrt{9x - 5} + (27x - 4)^{\frac{1}{3}}}$ is equal to
(A) 2
(B) 3
(C) 4
(D) 1

55. For the function $f(x) = x \sin \pi x$, $x > 0$, then number of solution of $f'(x) = 0$ in $[0, 2]$ is
(A) 2
(B) 3
(C) 4
(D) 1

56. If exhaustive values of x satisfying $|\cos^{-1} x| + |\sin^{-1} x| + |\tan^{-1} x| = |\pi - \cot^{-1} x|$ belongs to [a, b], then |a| + |b| is equal to
(A) 0
(B) 1
(C) 2
(D) 3

57. The number of solution of $\sin^2 2x = \cos^2 x$ in $[0, 2\pi]$ is
(A) 4
(B) 5
(C) 6
(D) 7

58. Let line 2x + y = 3 intersects a circle $x^2 + y^2 = 16$ at A and B. Now tangents are drawn to circle $x^2 + y^2 = 16$ at A and B which intersect at $P(\alpha, \beta)$, then $\alpha + \beta$ is equal to
(A) 8
(B) 12
(C) 16
(D) 18

59. Consider the cubic polynomial $f(x) = x^3 - ax^2 + bx + c$. If the equation f(x) = 0 has integral roots α, β, $\gamma \in [0, 5]$ such that f(5) = 2, then integral value of a is
(A) 8
(B) 9
(C) 11
(D) 12

60. If the line x + y = 1 touches an ellipse $\frac{x^2}{4} + \frac{y^2}{3} = \frac{1}{12}$ at $P(x_1, y_1)$, then $\frac{1}{3x_1 + 8y_1}$ is equal to
(A) 1
(B) $\frac{1}{2}$
(C) $\frac{1}{3}$
(D) $\frac{1}{4}$

61. Exhaustive set of values of x satisfying equation [[x] + 2x] = 4 is (where [.] denote greatest integer function)
(A) (1, 2)
(B) $\left(\frac{3}{2}, 2\right]$
(C) $\left[\frac{3}{2}, 2\right)$
(D) $\left[\frac{3}{2}, 3\right)$

62. $f : R \to Y,\ f(x) = \dfrac{e^x - e^{|x|}}{10e^x + e^{|x|}}$ is onto function, then Y is

(A) $\left(-\dfrac{1}{10},\ 0\right]$

(B) $\left(-\dfrac{1}{10},\ \infty\right)$

(C) (–1, 0]

(D) (–1, 1]

63. The number of critical points of f(x) = max(|tan x|, cos |x|) in $(-\pi, \pi)$ is

(A) 3

(B) 4

(C) 5

(D) 2

64. A tangent at P(5, 5) to hyperbola $\dfrac{(x-1)^2}{4} - \dfrac{(y-2)^2}{3} = 1$ meets its asymptotes at Q and R. If C be the centre of hyperbola, then area of triangle CQR is equal to

(A) 12

(B) 6

(C) $2\sqrt{3}$

(D) $3\sqrt{2}$

65. If a – b = 1, b – c = 2, a, b, c $\in$ R, then value of $|a + bw + cw^2|^2$ where w is cube root of unity ($w \neq 1$)

(A) 5

(B) 7

(C) 9

(D) none of these

66. Two lines $\dfrac{x-1}{2} = \dfrac{y+1}{3} = \dfrac{z-1}{4}$ and $\dfrac{x-3}{1} = \dfrac{y-k}{2} = \dfrac{z}{1}$ intersect at a point $P(\alpha, \beta, \gamma)$, then $\alpha + \gamma - 2\beta$ is equal to

(A) 6

(B) 4

(C) 5

(D) 7

67. $\sim (p \vee q) \vee (\sim p \wedge q)$ is logically equivalent to

(A) ~p

(B) p

(C) q

(D) ~q

68. A pair of dice is rolled till a sum of 4 or 7 is obtained. The probability that 7 comes before 4 is

(A) $\frac{2}{7}$

(B) $\frac{2}{3}$

(C) $\frac{1}{3}$

(D) $\frac{3}{5}$

69. Let $\vec{a}$ and $\vec{b}$ be unit vectors, $\vec{a}\cdot\vec{b}=0$ and if $\alpha=\left[\vec{a}+\vec{a}\times\vec{b},\ \vec{b}+\left(\vec{a}\times\vec{b}\right),\ \vec{a}\times\vec{b}\right]$,

$\beta=\left(\vec{a}\times\vec{i}\right)\cdot\left(\vec{b}\times i\right)+\left(\vec{a}\times\vec{j}\right)\cdot\left(\vec{b}\times\vec{j}\right)+\left(\vec{a}\times\vec{k}\right)\cdot\left(\vec{b}\times\vec{k}\right)$, then $\alpha+\beta$ is equal to

(A) 0
(B) 1
(C) 2
(D) none of these

70. If number of solution of equation $z^2=\bar{z}$ where z = x + iy is n, then the coefficient x^{n+7} in expansion of $(2x^2+x-3)^6$ is

(A) 190
(B) 192
(C) 194
(D) 196

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. If y = f(x) passing through (4, 2) satisfies the differential equation $y(1+xy)dx - xdy = 0$, then f(4) is equal to

72. If $\int\frac{5\sin x}{\sin x-2\cos x}dx=Ax+B\ln\left|\sin x-2\cos x\right|+c$, then 2A + B is equal to

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. In an experiment with 20 observations on x, the following results were available $\Sigma x^2=2530$, $\Sigma x=210$. On observation that 30 was found to be wrong and replaced by the correct value 40. Then the correct variance is

74. PQRS is a trapezium such that QR $\perp$ RS. If $\angle PSQ = 30°$, $QR=3\sqrt{3}$, RS = 5, then PQ is equal to

75. If $x=1+\frac{1}{4}+\frac{1}{4^2}+.....\infty$, $y=1+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+.....\infty$, $z=1+\frac{1}{c}+\frac{1}{c^2}+\frac{1}{c^3}+.....\infty$. If x, y, z are in H.P. and |c| > 1, then c + 3 is equal to